Experiment- 2 Connecting to Data Source, Connecting to Database, Different types of Tableau Joins.

Tableau Joins:

A **join** combines data from two (or more) tables based on a **related field** (key). In Tableau, joins happen **before** the visualization step — they are part of the data source definition.

2.1 Types of Joins in Tableau

Tableau supports the standard SQL join types:

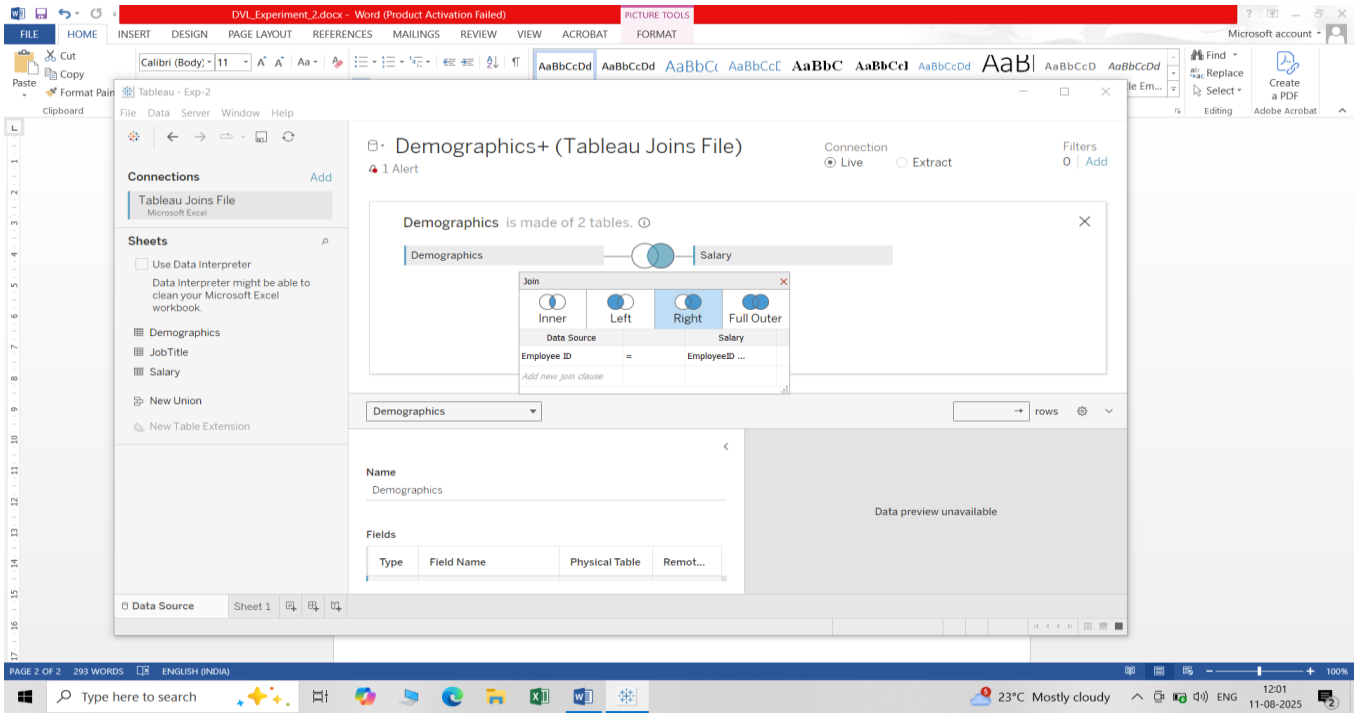
Join Type	Symbol in Tableau	Definition	Example Result
Inner Join	⊗ (two circles overlapping)	Returns **only** the rows that have matching values in both tables.	Students who **have** both ID in "Students" and "Exam Scores".
Left Join	⊙→ (left circle full, right circle partial)	Returns all rows from the **left table**, plus matching rows from the right table. Missing matches → NULLs.	All Students, even if they don't have exam scores.
Right Join	←⊙ (right circle full, left circle partial)	Returns all rows from the **right table**, plus matching rows from the left table. Missing matches → NULLs.	All Exam Scores, even if they don't match a student.
Full Outer Join	(both circles full)	Returns **all rows** from both tables, matching where possible. Non-matching → NULLs.	All Students and all Exam Scores, matching where IDs exist.

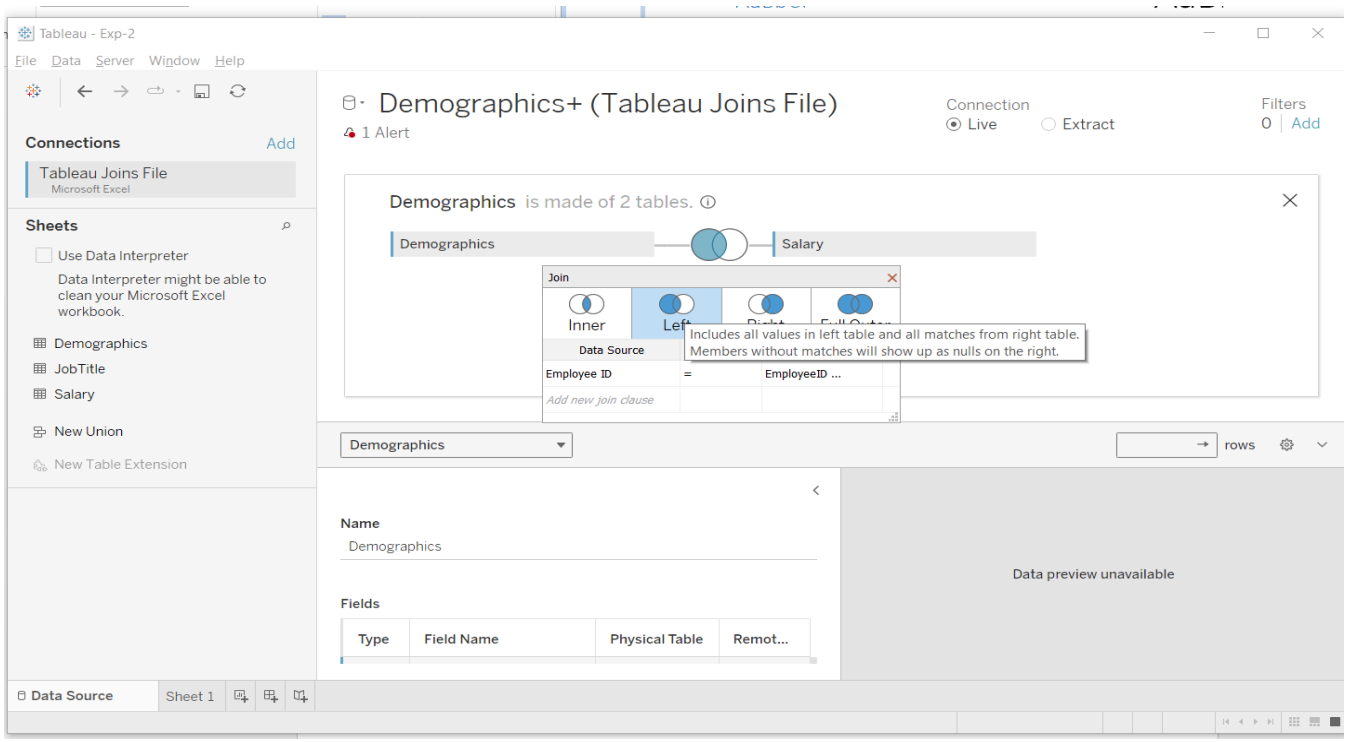
2.2 How to Create a Join in Tableau

1. **Drag the first table** into the Data Source canvas.
2. **Drag the second table** next to it.
3. Tableau will show a **join icon** between them.
4. Click the join icon to:
    - Select **join type** (Inner, Left, Right, Full Outer).
    - Define **join clause** → select matching fields (e.g., `table1.CustomerID = table2.CustomerID`).
    - Add multiple join conditions if required.

2.3 Join Clauses

- You can join on:
  - Same name fields (automatic)
  - Different name fields (manual mapping)
- You can use **calculated join clauses** (e.g., `UPPER(table1.City) = UPPER(table2.City)`).





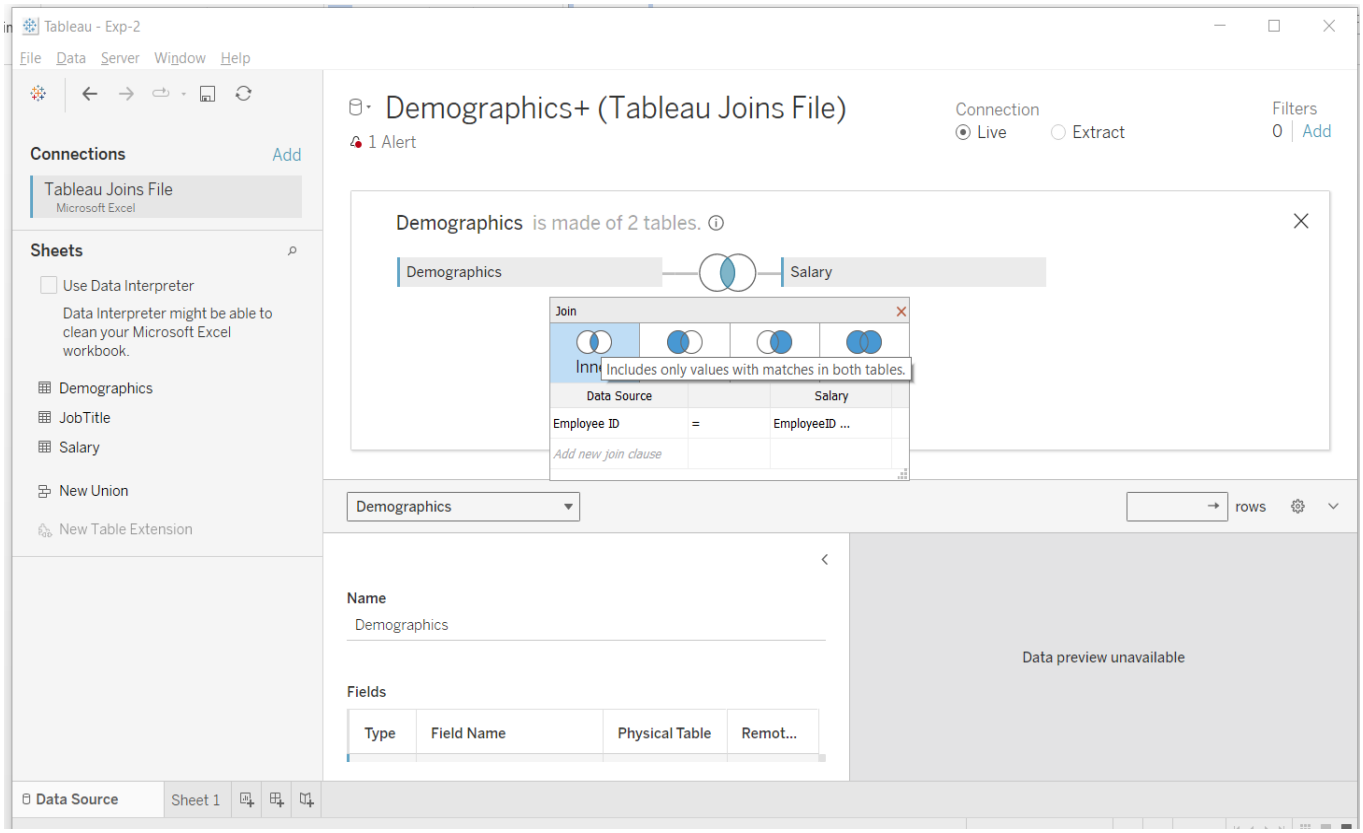


Tableau - Exp-2
File Data Server Window Help
Connections Add
Tableau Joins File
Microsoft Excel
Sheets
Use Data Interpreter
Data Interpreter might be able to clean your Microsoft Excel workbook.
Demographics
JobTitle
Salary
New Union
New Table Extension
Demographics+ (Tableau Joins File)
1 Alert
Connection
Live
Extract
Filters
0 Add
Demographics is made of 2 tables.
Demographics
Salary
Join
Includes only values with matches in both tables.
Data Source
Salary
Employee ID
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Add new join clause
Demographics
rows
Name
Demographics
Fields
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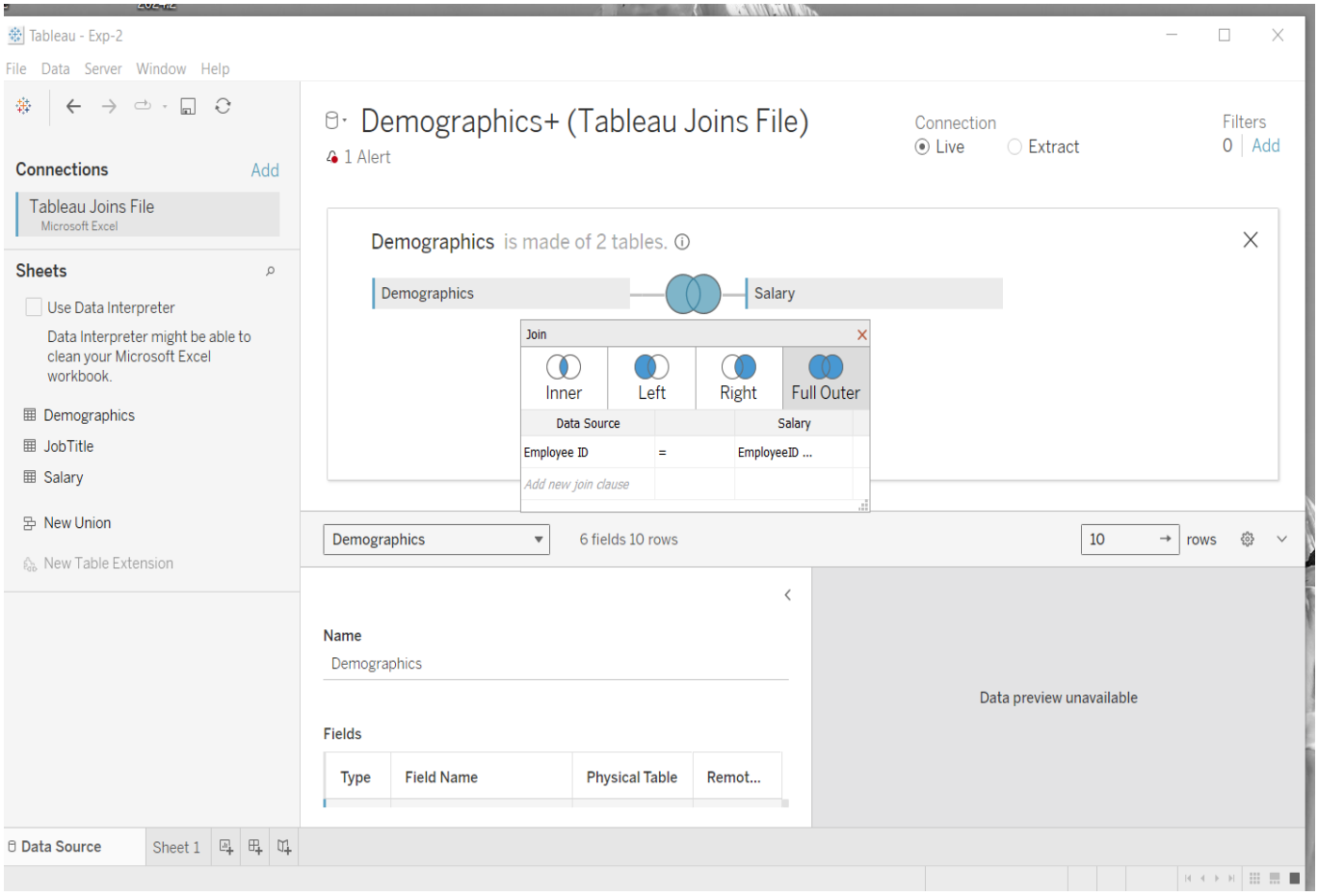
Field Name
Physical Table
Remot...
Data preview unavailable
Data Source
Sheet 1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